

Fatima Anwar

Ph. D. Chemistry

University of North Texas | Denton Texas USA | Dec 2023

Process Support Engineer III

Applied Materials Inc. | Dallas Texas USA

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Summary

- Resourceful and self-driven R&D scientist in semiconductor process engineering and tool technology development.
- Hands-on experience with thin film deposition and plasma-etching: CVD, ALD, PVD, MBE and RIE.
- Proven leadership experience including heading project team resulting in NSF Grant (\$500,000), initiating UNT-AMAT research collaboration, mentoring undergraduates, and managing vendor interactions for AMAT processing equipment.

Work Experience

Applied Materials Inc.

Process Support Engineer III

Nov 2023 – Present

- Joined a team of Engineers working for the process support of Implant tools at customer site.
- Performed root cause analysis, troubleshooted complex hardware related issues, and modified tools for experimental and customer needs.
- Work with Business unit to communicate complex tool issues that customer face.

R&D Process Engineer Intern

Aug 2022 – July 2023

- Joined a team of scientists, engineers, and technicians for the development of a novel non-normal RIE technology for process fabrication of engineered optics components.
- Prepared optical and semiconductor samples, executed design of experiments, analyzed data with multiple variables, created automation tools for ease of large data handling and analysis of samples.
- Presented analysis, results, conclusions and presented product improvements in review meetings.
- Performed root cause analysis, troubleshooted complex hardware related issues, and modified tools for experimental and customer needs.
- Gained familiarity with AMAT technologies such as Sculpta Directional Etch, Ion Implantation, Sym 3 RIE and plasma doping (PLAD) tools and processes.
- Founded UNT-AMAT collaboration to solve technical challenges in new materials (Si, SiGe, Ge) modification for DRAM and logic devices.

Research Experience

Department of Chemistry | University of North Texas | Denton TX

Graduate Research Assistant | Thin Films and Surface Science Research Lab

Aug 2017 – Dec 2023

Thesis Topic: Surface Engineering of Materials for Beyond-Microelectronics | Adviser: Dr. Jeffry Kelber

- Developed thin film deposition processes of transition metal oxide, nitride, and carbide.
- Studied surface chemistry of films and interfacial interactions for spintronic devices and UHV-EC of Nitrogen reduction reactions.
- Developed Ultra-high Vacuum (UHV) systems for film deposition and installed multiple characterization capabilities.
- Mentored two undergrads and three junior graduate students through small defined research projects.
- Instructed General Chemistry labs for undergraduate students.

Selected Project I: MBE growth of ultrathin chromia on a hexagonally ordered d^0 ferromagnet: Evidence of interfacial exchange bias at the $\text{Cr}_2\text{O}_3/\text{TiO}_{2-x}$ Interface.

- Developed e-beam evaporation (MBE) method for successful deposition of $\sim 10 \text{ \AA}$ Cr_2O_3 (0001) film on $\sim 50 \text{ \AA}$ TiO_{2-x} (111) film heterostructure on Al_2O_3 (0001).
- Determined interfacial heterostructure chemistry using XPS, EELS, and LEED data as an evidence of Cr deposition yielding increased Ti (III) formation in the titania substrate and the formation of a Cr_2O_3 overlayer, without Cr/Ti interfacial mixing.
- Results suggested a straightforward route toward the formation of chemically and magnetically robust, voltage-switchable spintronic devices – verified through measurement of interfacial exchange bias after extensive ambient exposure at RT.

Selected Project II: Atomic layer deposition of BN as protective coating of thermochromic Vanadium (IV) oxide for spacecraft application

- Led the team for developing MBE and magnetron sputter deposition processes for tuning thermochromic (VO_2) film composition for spacecraft applications.
- Developed e-beam evaporation (MBE) and atomic layer deposition (ALD) method for successful deposition of $\sim 300 \text{ \AA}$ VO_2 film and $\sim 3 \text{ nm}$ BN film, respectively.
- Investigated the role of BN as protective coating against degradation of VO_2 by atomic oxygen attack and UV irradiation through XPS.
- Resulted in understanding the interfacial chemistry of films and surface interactions with low earth orbit environment

Education

- Ph.D. Chemistry | CGPA 4.00 | Expected 2023 | University of North Texas | Denton TX USA
- MS Analytical Chemistry | CGPA 3.98 | 2016 | University of the Punjab | Lahore Pakistan
- BSc. Hons. Chemistry | CGPA 3.59 | 2014 | Government College University | Lahore Pakistan

Techniques and Technical Skills

CVD	PVD	XPS	Ellipsometry	Plasma-Etching
MBE	RIE	UPS	SEM	Python (Basic Knowledge)
ALD	AES	EELS	AFM	JMP (Basic Knowledge)
OES	LEED	XRD	Origin Pro	Macro Building

Awards and Honors

- Achieved *Team Award / Market share* at customer site (Texas Instruments) | Dec 2024 | Applied Materials Inc.
- Achieved *COS Dean's Doctoral Fellowship Fund* | Summer 2022 | Ph. D. Chemistry | UNT
- Achieved *R.B. Escue Chemistry Education Research Scholarship* | 2020-2021 | Ph. D. Chemistry | UNT
- Achieved *John A. Altermann and Katherine M Bradford Scholarship* | 2020-2021 | Ph. D. Chemistry | UNT
- Achieved *TPEG grant* | 2019-2020 | Ph. D. Chemistry | UNT
- Achieved *Porter Evans Scholarship* | 2019-2020 | Ph. D. Chemistry | UNT
- Achieved *Distinction-2nd Position* | MS. Analytical Chemistry | PU
- Achieved *Distinction-2nd Position* | BS Hons. Chemistry | GCU

Presentation

- Presented industrial improvement for Ion Implantation Technology of Applied Materials Inc. at Texas Instruments at Annual GATE Conference organized by Applied Materials Inc. in Hanoi, Vietnam | Oct 2024

Publications

1. Ladewig C, Anwar F, Lee V, Kelber JA, Shah SQ, Dowben PA. Ultrathin Chromia on a Hexagonally-Ordered d0 Ferromagnet: Evidence of Interfacial Exchange Bias at the Cr2O3/TiO2-x Interface. *Langmuir*. 2019 Oct 29;35(46):14797- 803.
2. Adaeze Osonkie, Ashwin Ganesan, Precious Chukwunye, Fatima Anwar, Kabirat Balogun, Mojgan Gharaee, Ishika Rashed, Thomas R. Cundari, Francis D'Souza, and Jeffry A. Kelber. Electrocatalytic Reduction of Nitrogen to Ammonia: the Roles of Lattice O and N in Reduction at Vanadium Oxynitride Surfaces. *ACS Applied Materials & Interfaces* 2022 14 (1), 531-542.
3. Chukwunye P, Ganesan A, Gharaee M, Balogun K, Anwar F, Adesope Q, Cundari TR, D'Souza F, Kelber JA. Electrocatalytic selectivity for nitrogen reduction vs. hydrogen evolution: a comparison of vanadium and cobalt oxynitrides at different pH values. *Journal of Materials Chemistry A*. 2022;10(40):21401-15.
4. Balogun, K., Precious Chukwunye, Fatima Anwar, Ashwin Ganesan, Qasim Adesope, Dominic Willadsen, S. Nemšák et al. "Interaction of molecular nitrogen with vanadium oxide in the absence and presence of water vapor at room temperature: Near -ambient pressure XPS." *The Journal of Chemical Physics* 157, no. 10 (2022): 104701.
5. Samar Alhowity, Kabirat Balogun, Ashwin Ganesan, Colton J. Lund, Olatomide Omolere, Qasim Adesope, Precious Chukwunye, Stella C. Amagbor, Fatima Anwar, M. K. Altafi, Francis D'Souza, Thomas R. Cundari, and Jeffry A. Niobium Carbide and Tantalum Carbide as Nitrogen Reduction Electrocatalysts: Catalytic Activity, Carbophilicity, and the Importance of Intermediate Oxidation States. *ACS Applied Materials & Interfaces* 2024 16 (2), 2180-2192

Affiliations & Hobbies

- Peer Reviewer for ECS Journal of Solid State Science and Technology
- Fellow member of American Institute of Chemistry (AIC)
- Full member of SigmaXi The Scientific Research Honor Society
- Mentor for New College Graduates in industry Applied Materials Inc.
- Excellent cook who likes adventurous travelling and spend holidays with family and / in exploring new places.